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Session 2: Training intensity distribution for endurance performance

Presentation · October 2017

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Training intensity, volume and recovery distribution among elite endurance athletes [View project](#)

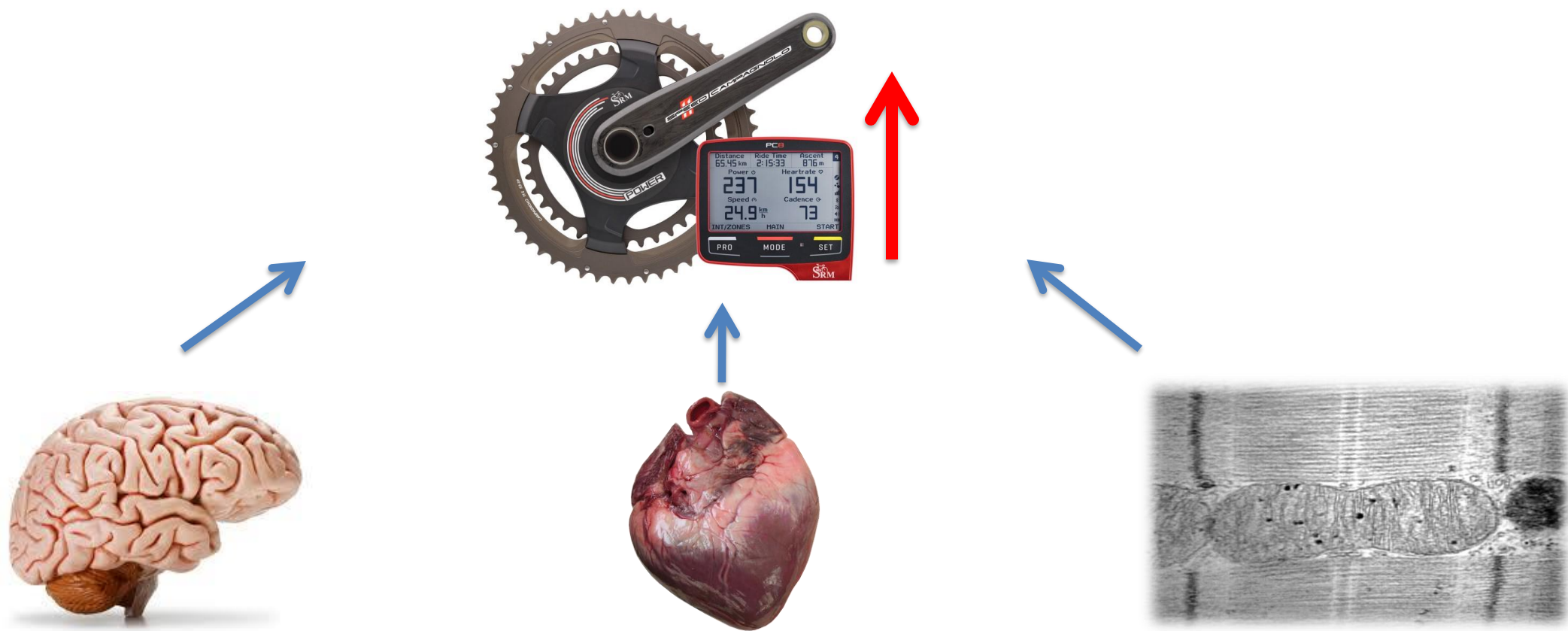


Session 2: **Training intensity distribution** for endurance athletes

From *polarized* to *optimized*?
Moving towards 2025

Stephen Seiler
University of Agder
Kristiansand, Norway





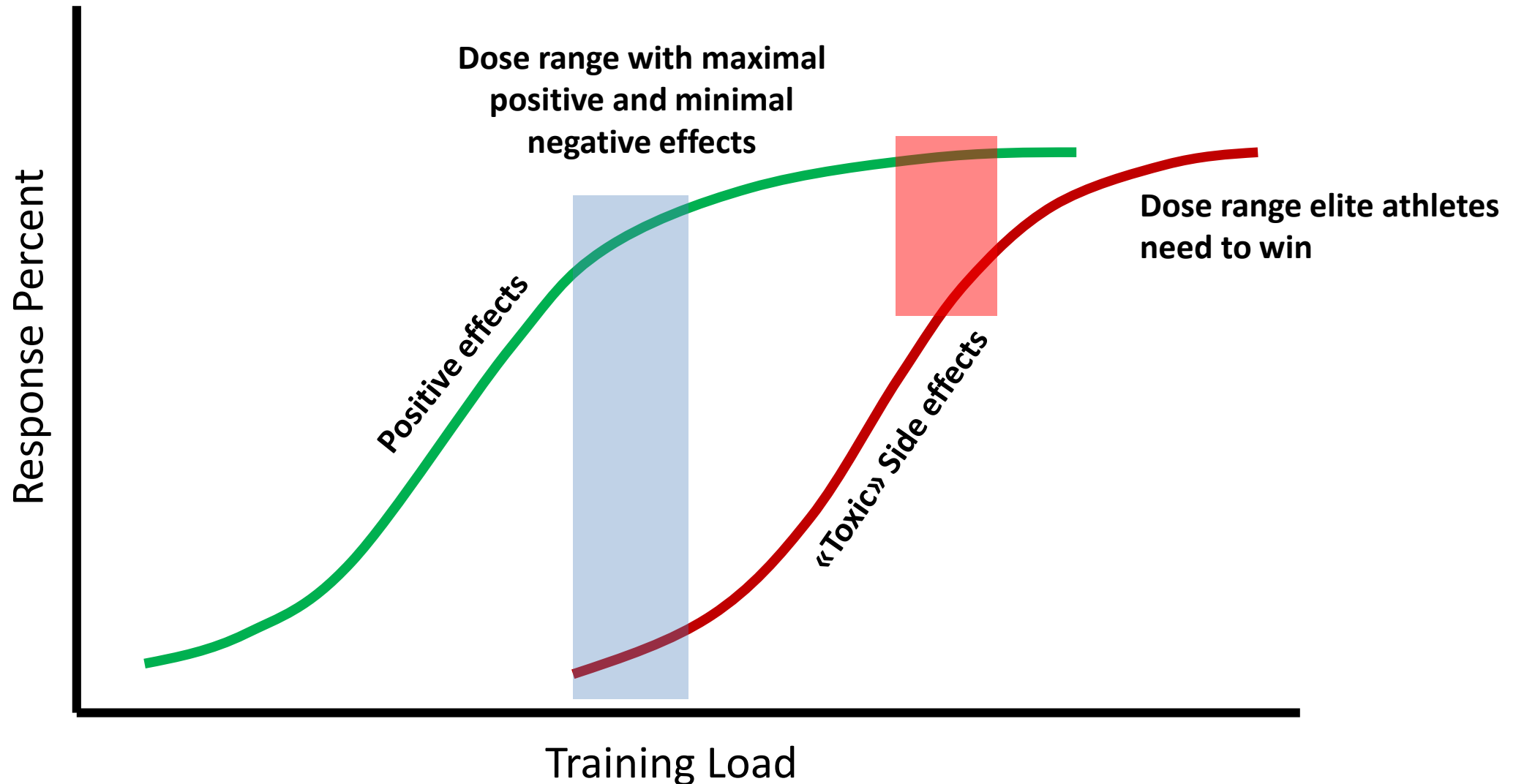
Adaptive Stimulus

Stress



(Adjusting training characteristics)

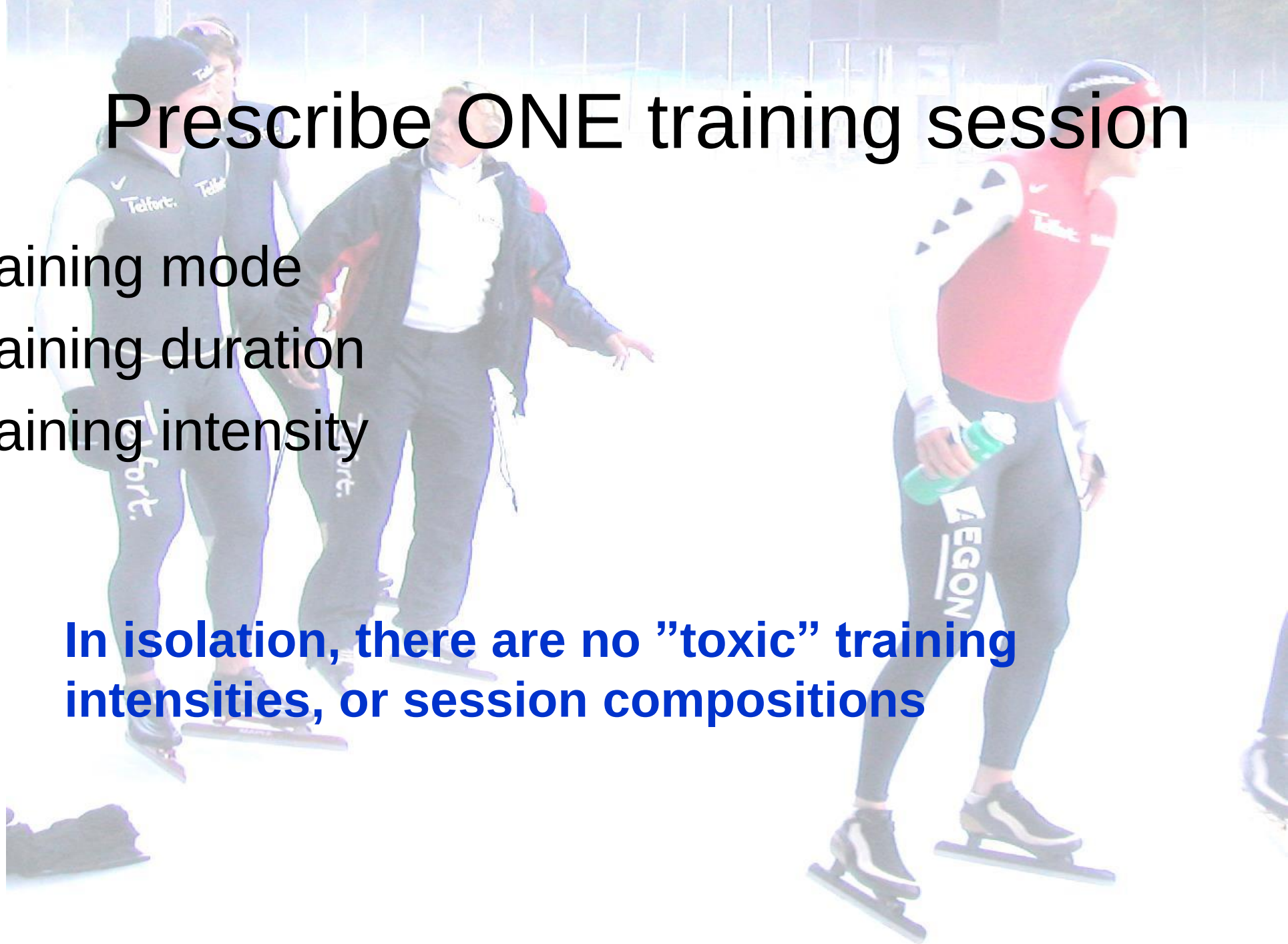
The right amount of a good thing?- Dose-Response Relationship



Prescribe ONE training session

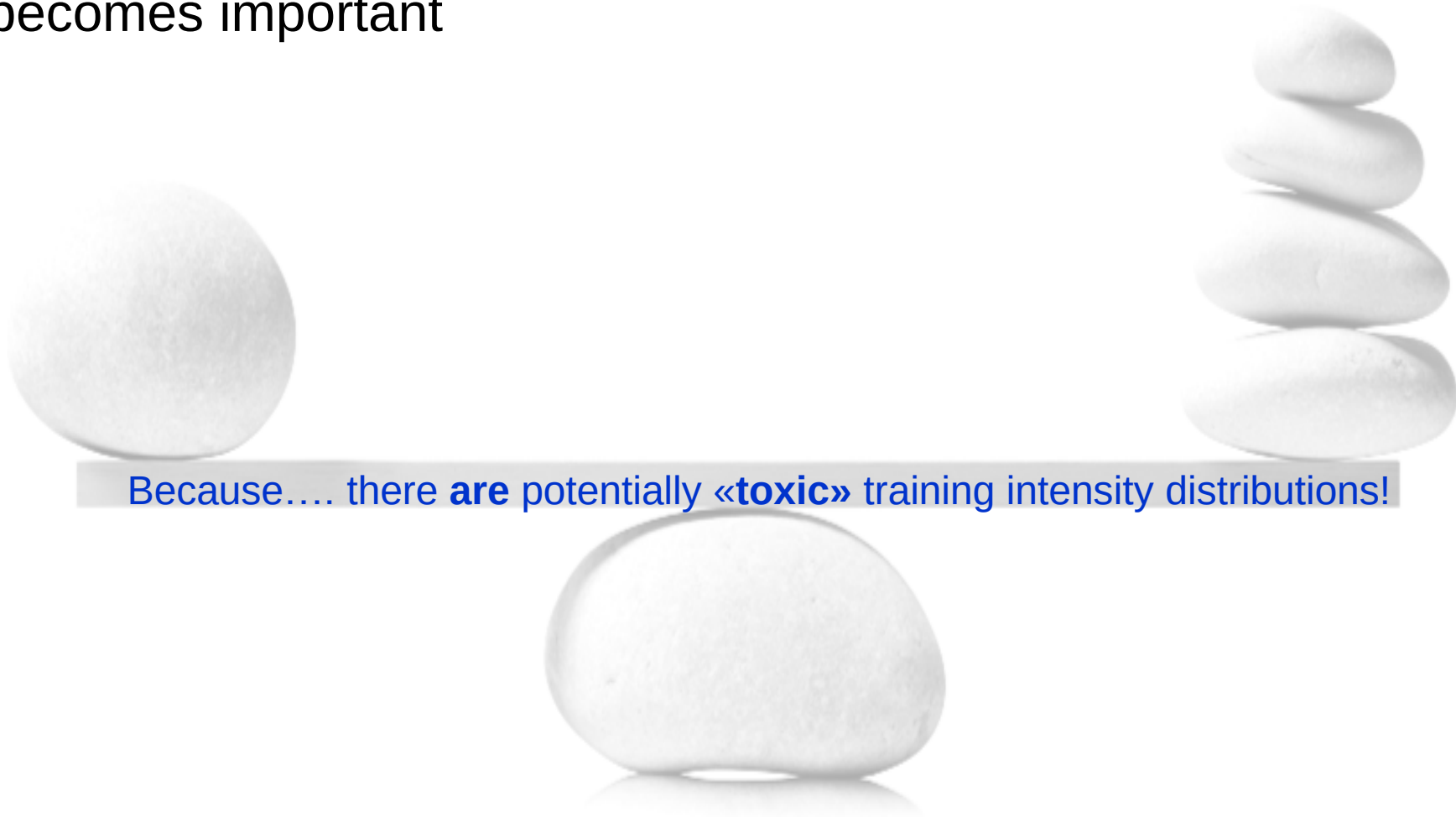
- Training mode
- Training duration
- Training intensity

In isolation, there are no "toxic" training intensities, or session compositions



Prescribe ONE (typical) WEEK

Balancing **stress** load and **stimuli** magnitude – intensity distribution and volume becomes important

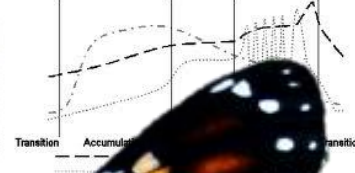
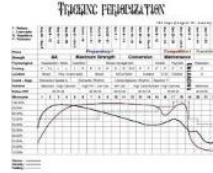


Because.... there **are** potentially «toxic» training intensity distributions!

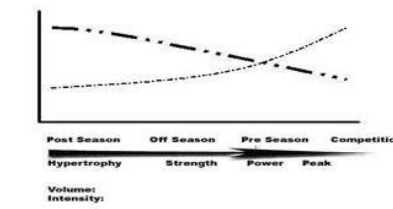
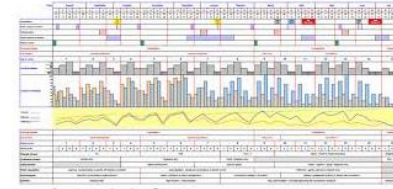
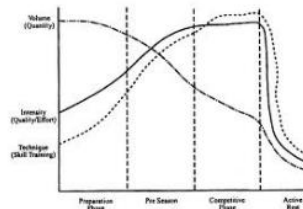
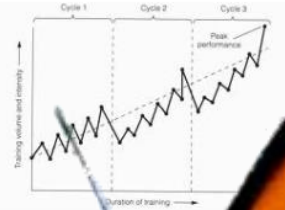
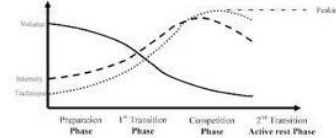
Prescribe 12 (building) WEEKS

Period	Purpose	Under Age 20	Over Age 20	Planned Recovery
Prep	Prepare to train—cross training, weights, functional strength, general conditioning	2-6 weeks	2-6 weeks	N/A
Base 1	Endurance, Force (weight), Speed Skill	4 weeks	3 weeks	Last 4-7 days
Base 2	Endurance, Force (skill), Speed Skill, limited muscular endurance	4 weeks	3 weeks	Last 4-7 days
Build 1	Endurance, Force (skill), Speed Skill, more muscular endurance	4 weeks	3 weeks	Last 4-7 days
Build 2	Endurance, Force (skill), Speed Skill, more muscular endurance, more specificity muscular and/or endurance Endurance, Power (skill), Maximal Aerobic Endurance, Force, Speed Skill, Break (skill)	4 weeks	3 weeks	Last 4-7 days
Peak	Simulate a portion of the race every 72 hours. Off-race, active recovery. Peak, stay sharp with 1-3 mile interval at race effort (70-75) decreasing in number daily.	1-2 weeks	1-2 weeks	N/A
Race		1 week	1 week	N/A
Taper	Peak and active recovery	1-6 weeks	1-6 weeks	N/A

Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100



Date	Week 1-3	Week 4-6	Week 7	Week 8-10
Aerobic	Base endurance Steady pace	Lactic acid tolerance training Intervals faster & slower		Aerobic power Intervals fast & slow
Resistance training	15 min 20 min 25 min	30:30 60:30 90:30	Recovery	2X 4 min 2X 4 min Strength training
Functional training	15 min 10 min 5 min	12 min 5 min 6 min		5 min 3 min
Rest	60s 60s 60s	30s 60s 90s		180s 180s
Core	Base 30s 60s 90s	Endurance 90s 120s 150s		Strength 45s 30s



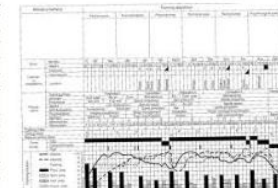
Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
General Preparation (off season)																					
Specific Preparation (pre season)																					
Pre-competition (late pre season)																					
Competition (competition)																					
Phase I: Utility and Mobility																					
Phase II: Load (endurance)																					
Phase III: Load (strength)																					
Phase IV: Performance (skill)																					
Competition (competition)																					
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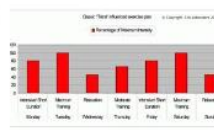
Week	Mon (Easy)	Tues (Hard)	Wed (Hard)	Thurs (Hard)	Fri (Easy)	Sat (Hard)	Sun (Hard)	Total
Week 1	4	4:00:00	5	5 Tempo	4	10	5	25
Week 2	4	4:00:00	5	5 Tempo	4	11	5	27
Week 3	4	4:00:00	5	5 Tempo	4	12	5	29
Week 4	4	4:00:00	5	5 Tempo	4	14	5	31
Week 5	4	4:00:00	5	5 Tempo	4	14	5	31

Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
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Competition (competition)																					



How much “control” over the basic training outcome do we have at the individual level?

Individual variation that is «genetic» and **non-responsive** to alterations in training prescription

The bigger **this** box, the less fun our job!

Individual variation that is **responsive** to adjustments in training prescription

Variation is large, but how much of it can be «corrected»?

Generalizable effects related to **periodization** characteristics **within** a best practice framework?

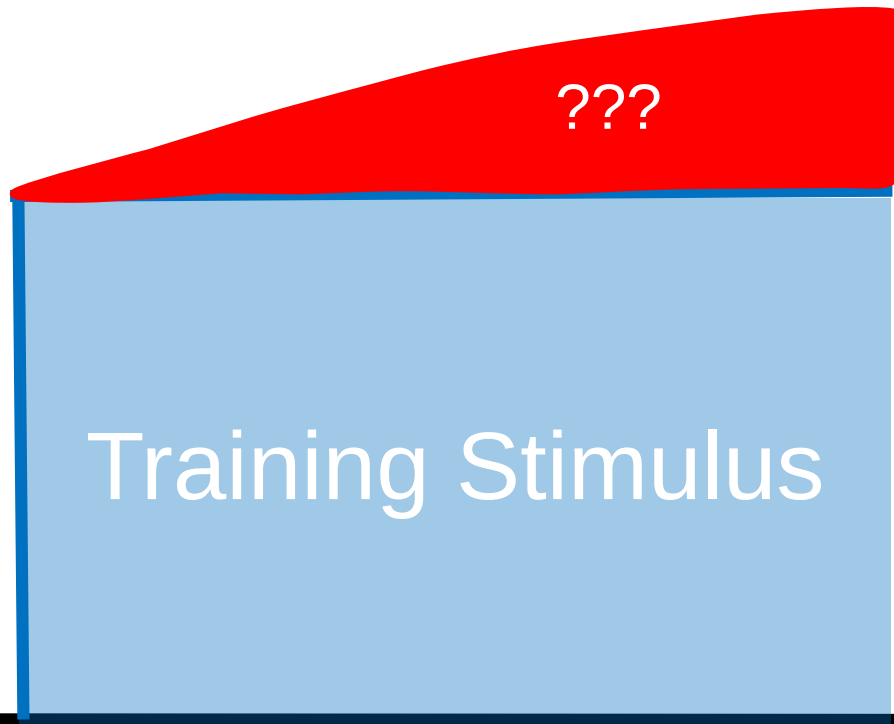
Small effects?- reality may not live up to the fancy lingo

Generalizable effects due to **basic framework** of intensity distribution and volume load?

15 years of work suggests this is **important!**

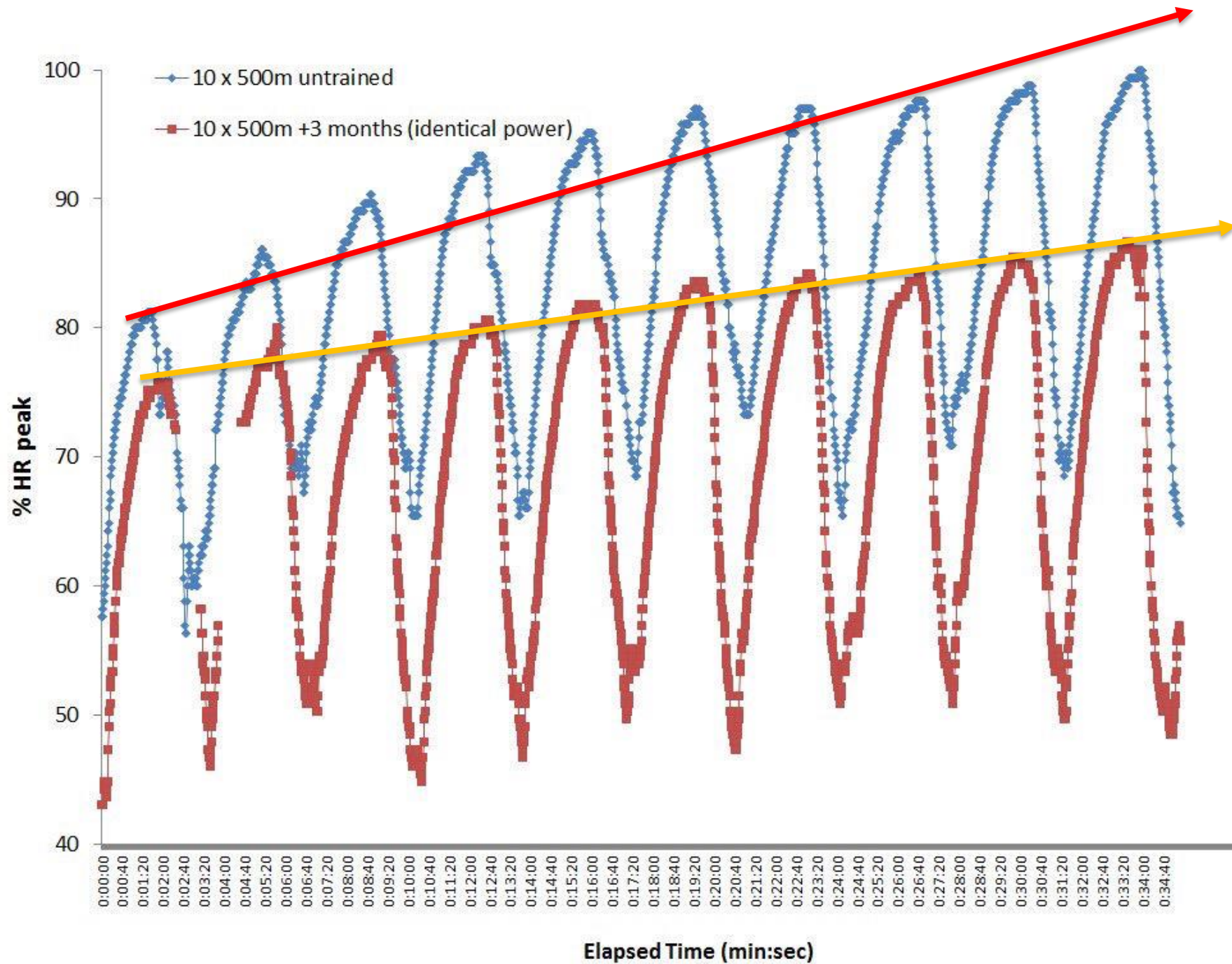
Acute «Dose» $\approx$ Intensity x Duration (but not exactly)

Intensity



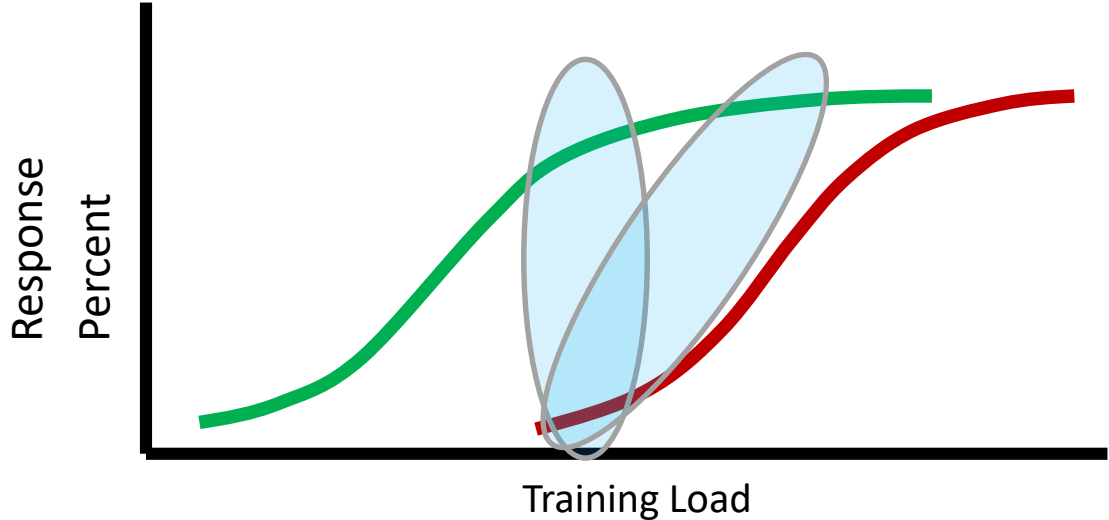
Individual response variation
and **situational** response
variation are both crucially
important!

Duration

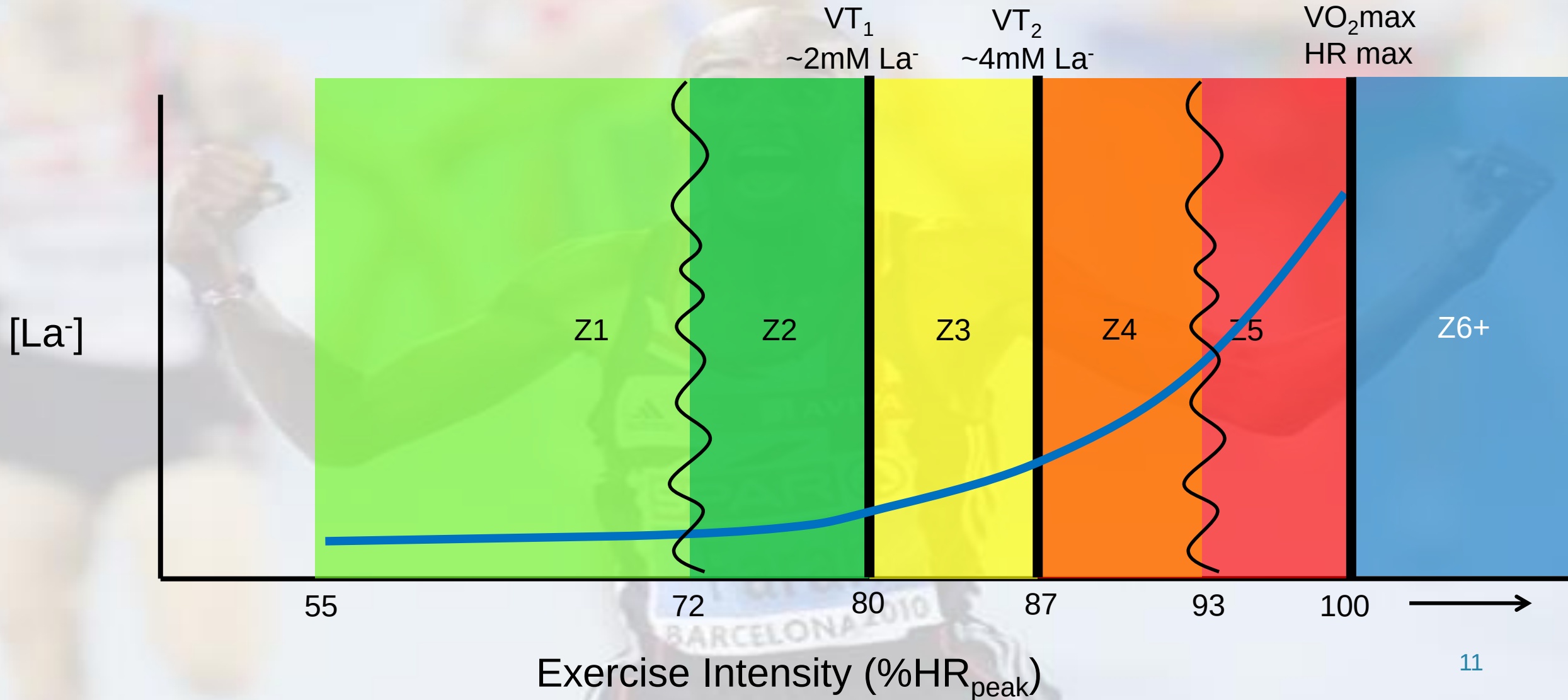


«Chronic» Dose-Response Risk/Benefit $\approx$ Training Volume x Intensity Distribution

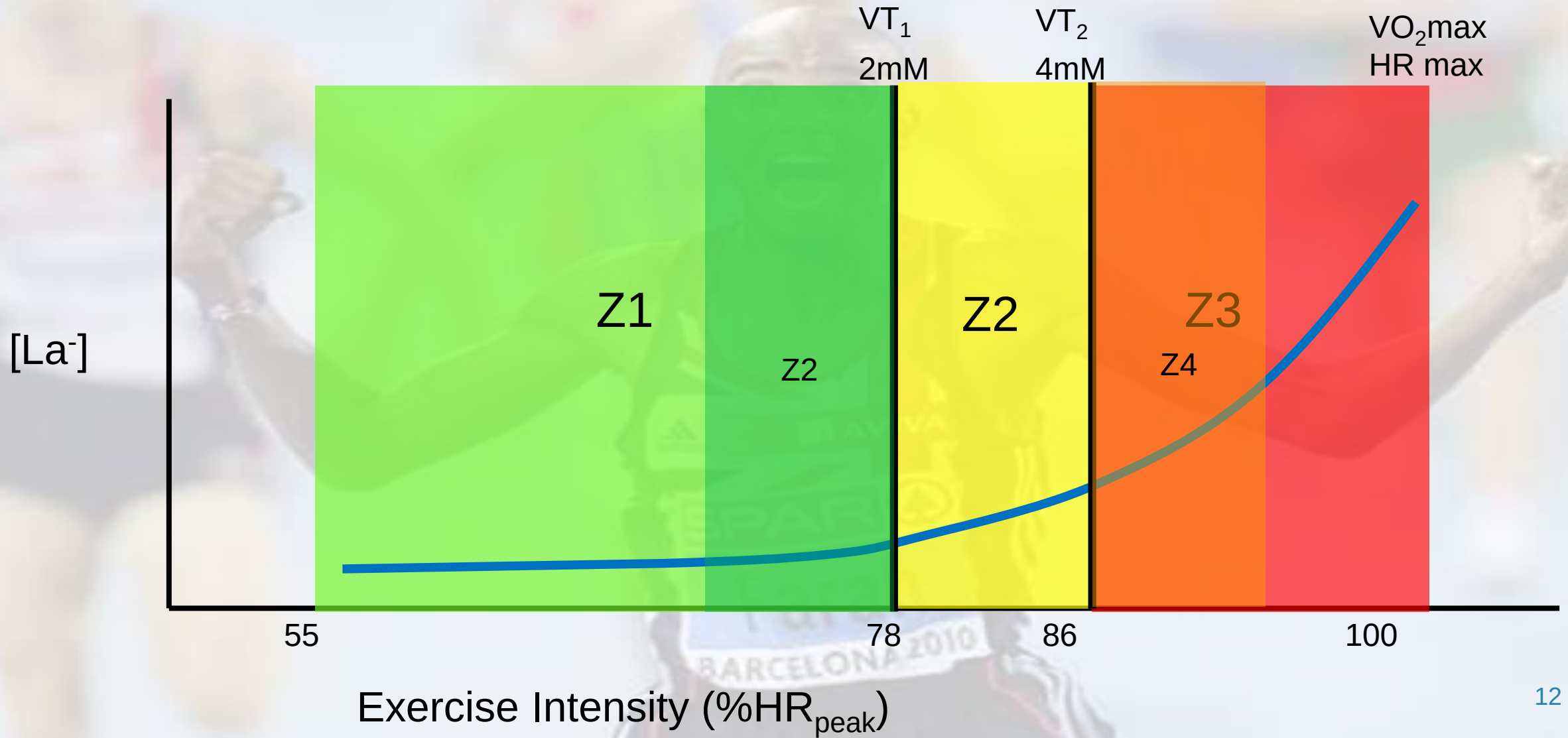
M	T	W	T	F	S	S	HIT	Th	LIT
LI	LI	Rest	LI	HI	Th	Rest	3	1	7
HI	LI	HI	LI	Rest	LI	LI			



A typical intensity zone scheme



3 intensity zones?



Intensity
zones:

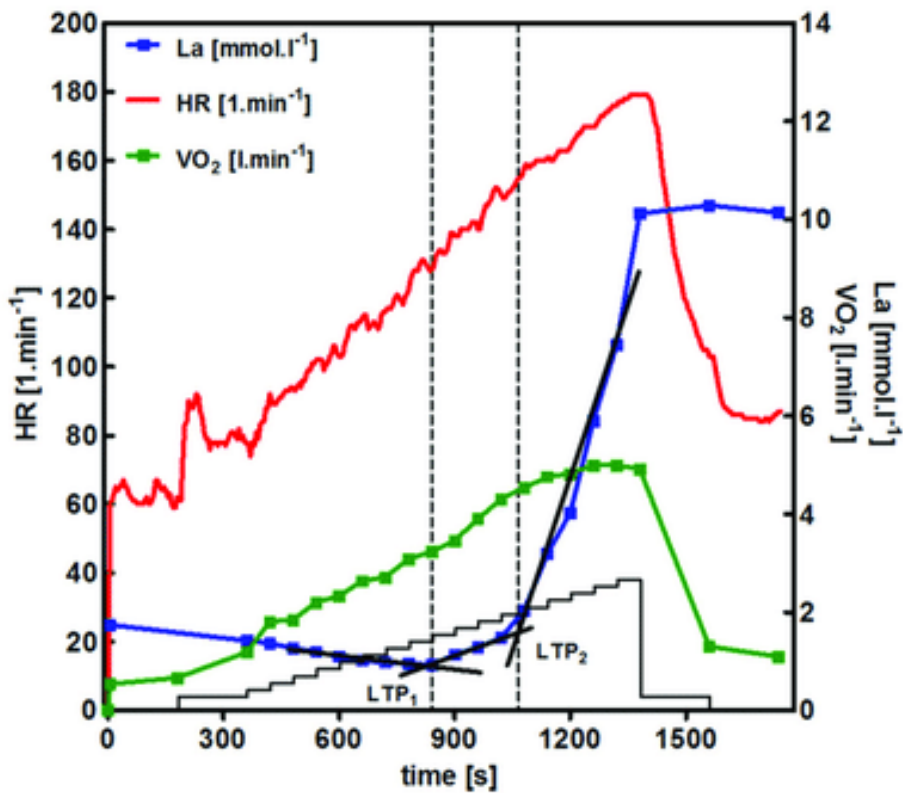
1

2

3

4

5



LTP₁:
P: 197 W (51.8% P_{max})
VO₂: 3.20 l.min⁻¹ (64.0% VO_{2max})
HR: 129 min⁻¹
La: 1.5 mmol.l⁻¹

LTP₂:
P: 284 W (74.7% P_{max})
VO₂: 4.59 l.min⁻¹ (91.8% VO_{2max})
HR: 159 min⁻¹
La: 2.5 mmol.l⁻¹

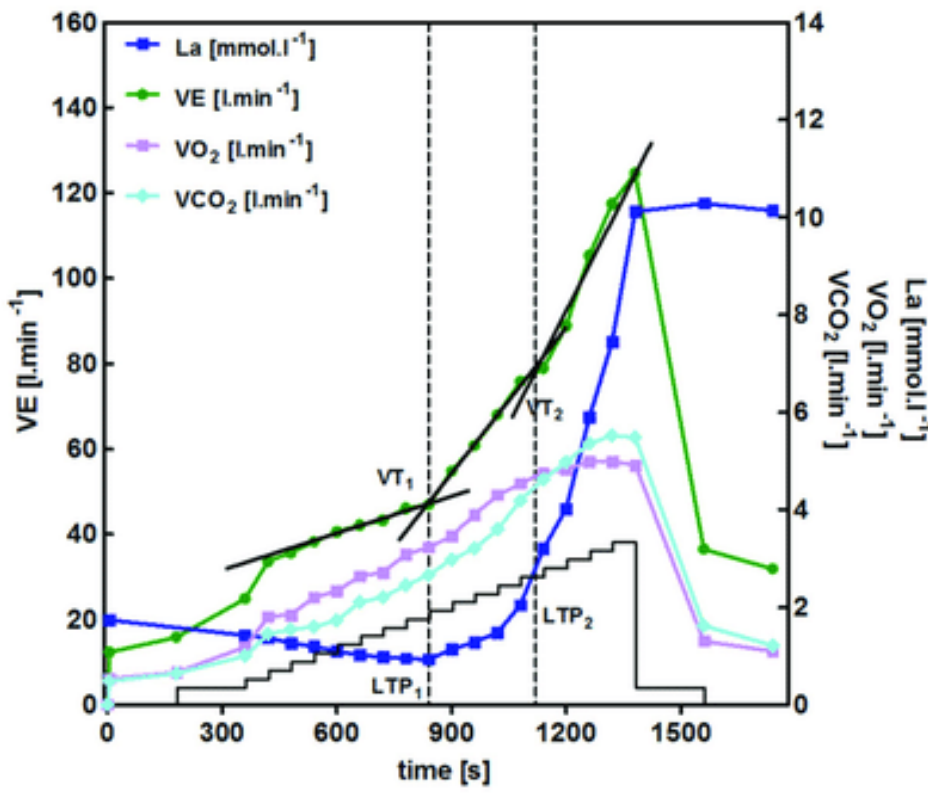
1

2

3

4

5



VT₁:
P: 200 W (52.6% P_{max})
VO₂: 3.23 l.min⁻¹ (64.6% VO_{2max})
HR: 130 min⁻¹
La: 1.5 mmol.l⁻¹

VT₂:
P: 282 W (74.2% P_{max})
VO₂: 4.57 l.min⁻¹ (91.4% VO_{2max})
HR: 159 min⁻¹
La: 2.4 mmol.l⁻¹

Figure from Hoffman & Tschakert, Frontiers in Physiology, May, 2017: https://www.researchgate.net/publication/317490542_Intensity-and_Duration-Based_Options_to_Regulate_Endurance_Training

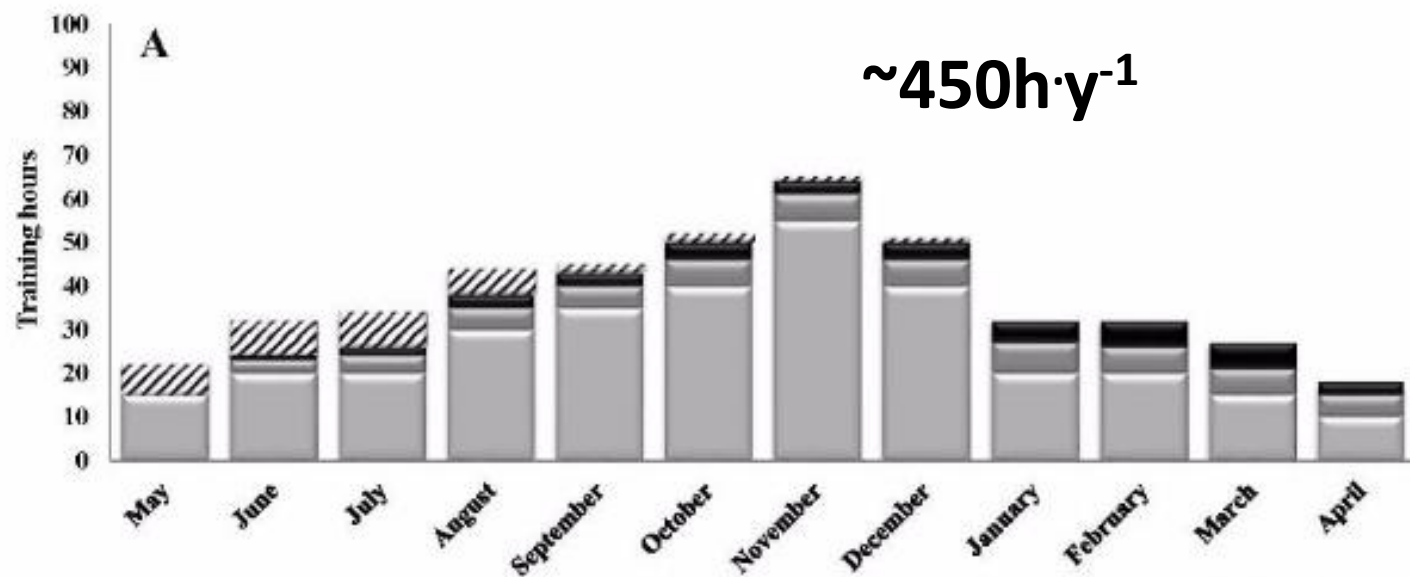
6 month training comparison

World Class (n=8)

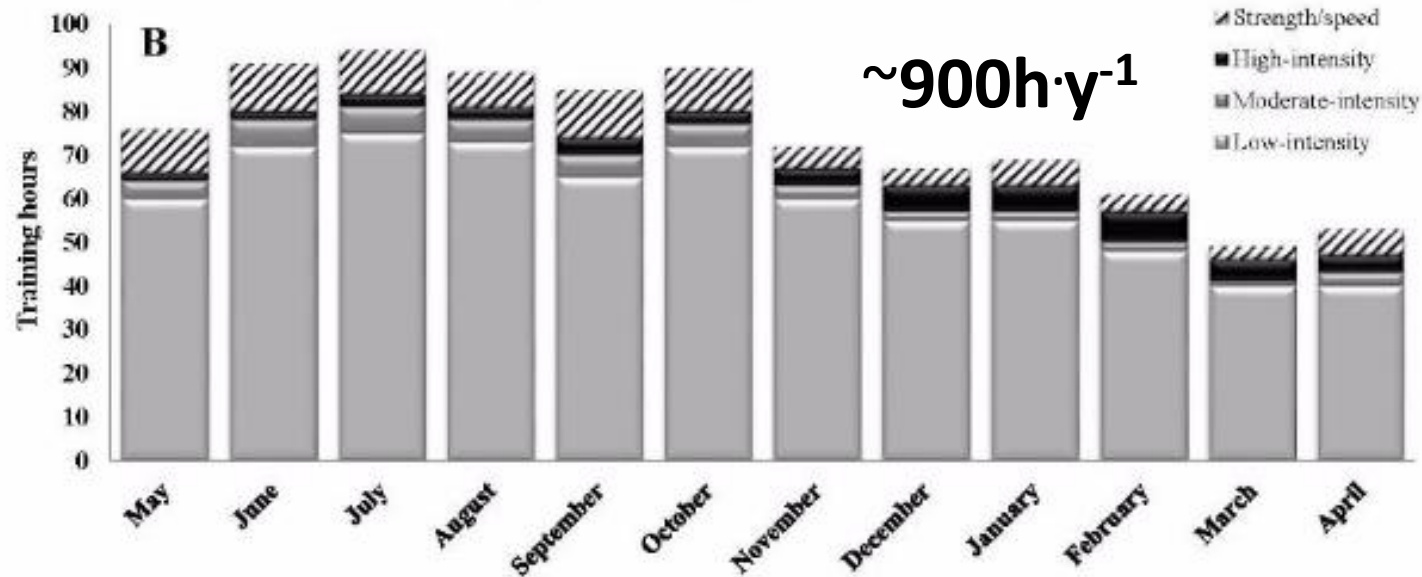
National Class (n=8)

	Training hours	% Total Training	Training hours	% Total Training
LIT	340 ± 23**	76.4 ± 4.6	254 ± 94	73.1 ± 12
MIT	29 ± 12**	6.5 ± 2.2	14 ± 6	4.4 ± 2.4
HIT	19 ± 3	4.4 ± 0.8	19 ± 8	5.6 ± 2.1
Speed	16 ± 7**	3.7 ± 1.5	7 ± 3	2.3 ± 1.2
Strength	39 ± 14	8.8 ± 2.9	31 ± 14	9.4 ± 3.7
Total	445 ± 27**	100	341 ± 90	100

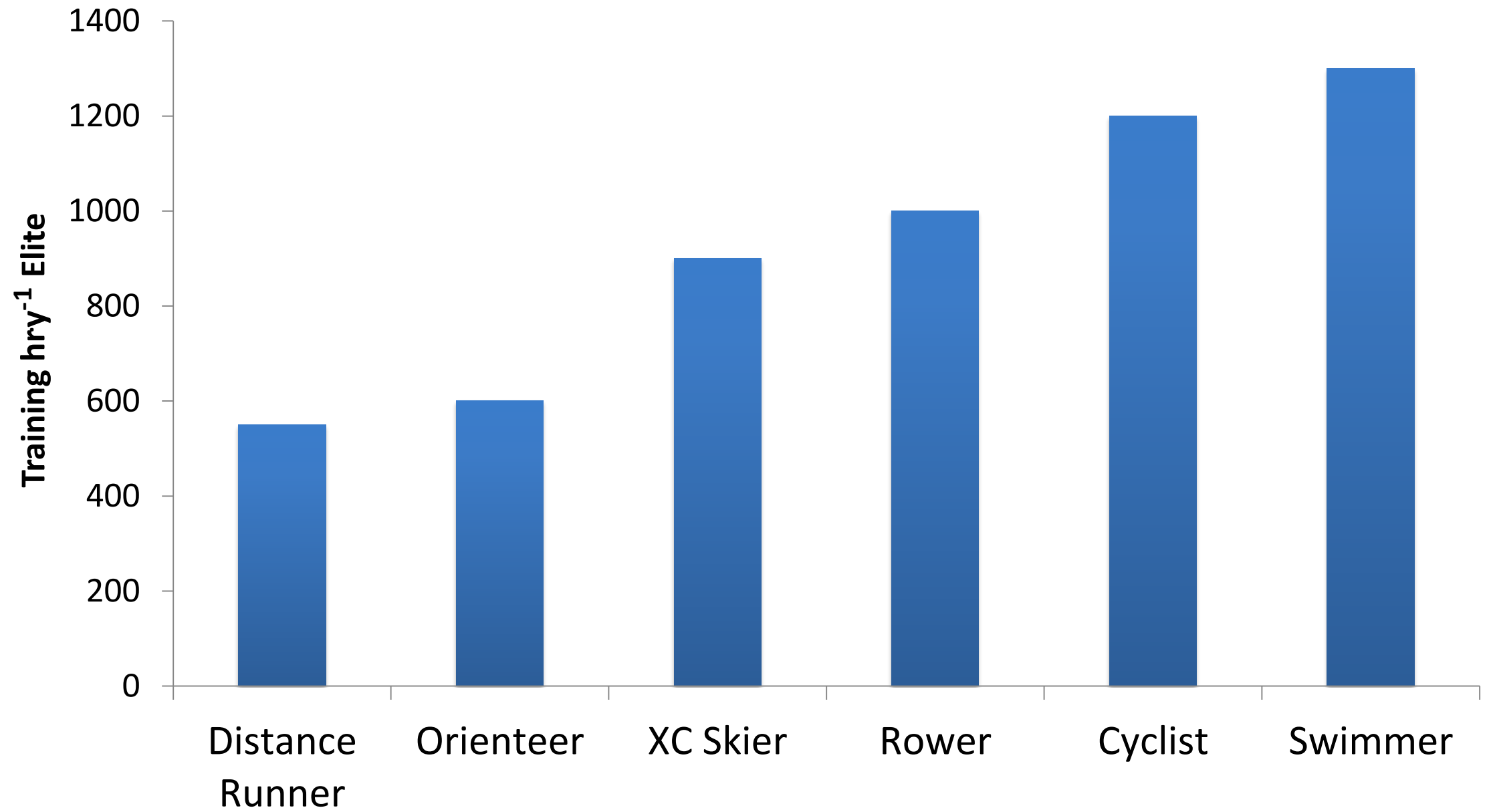
Sandbakk Ø, Holmberg HC, Leirdal S, Ettema G. The Physiology of World Class Sprint Skiers. *Scand J Med Sci Sports*. 2011 Dec;21(6):e9-16



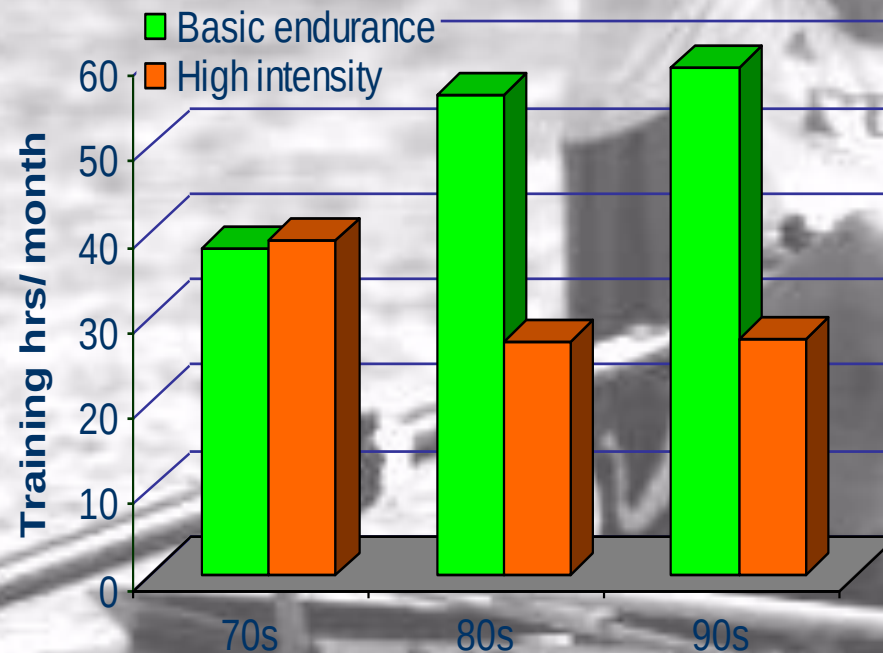
1965



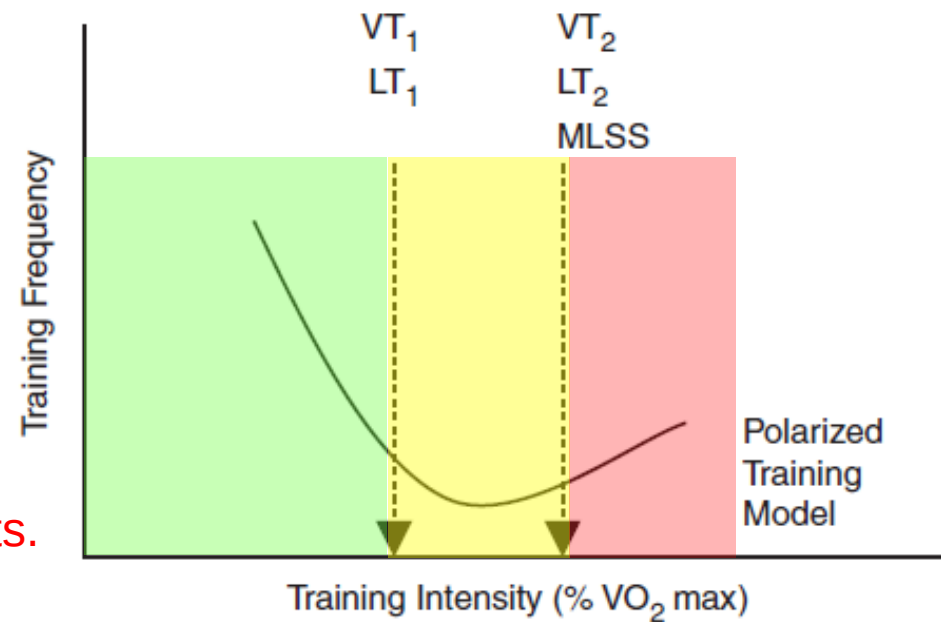
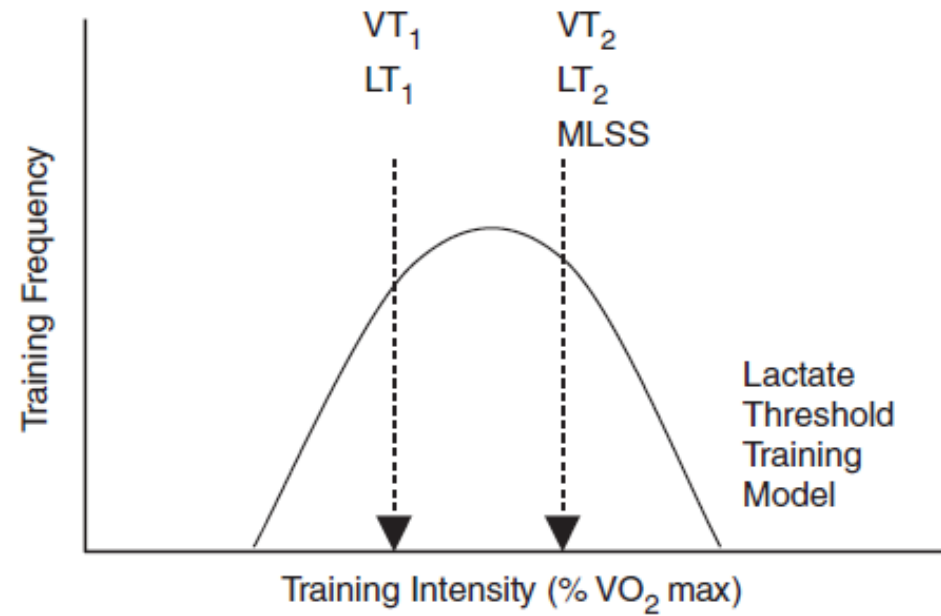
2015



Training Intensity Distribution International Rowing Medalists

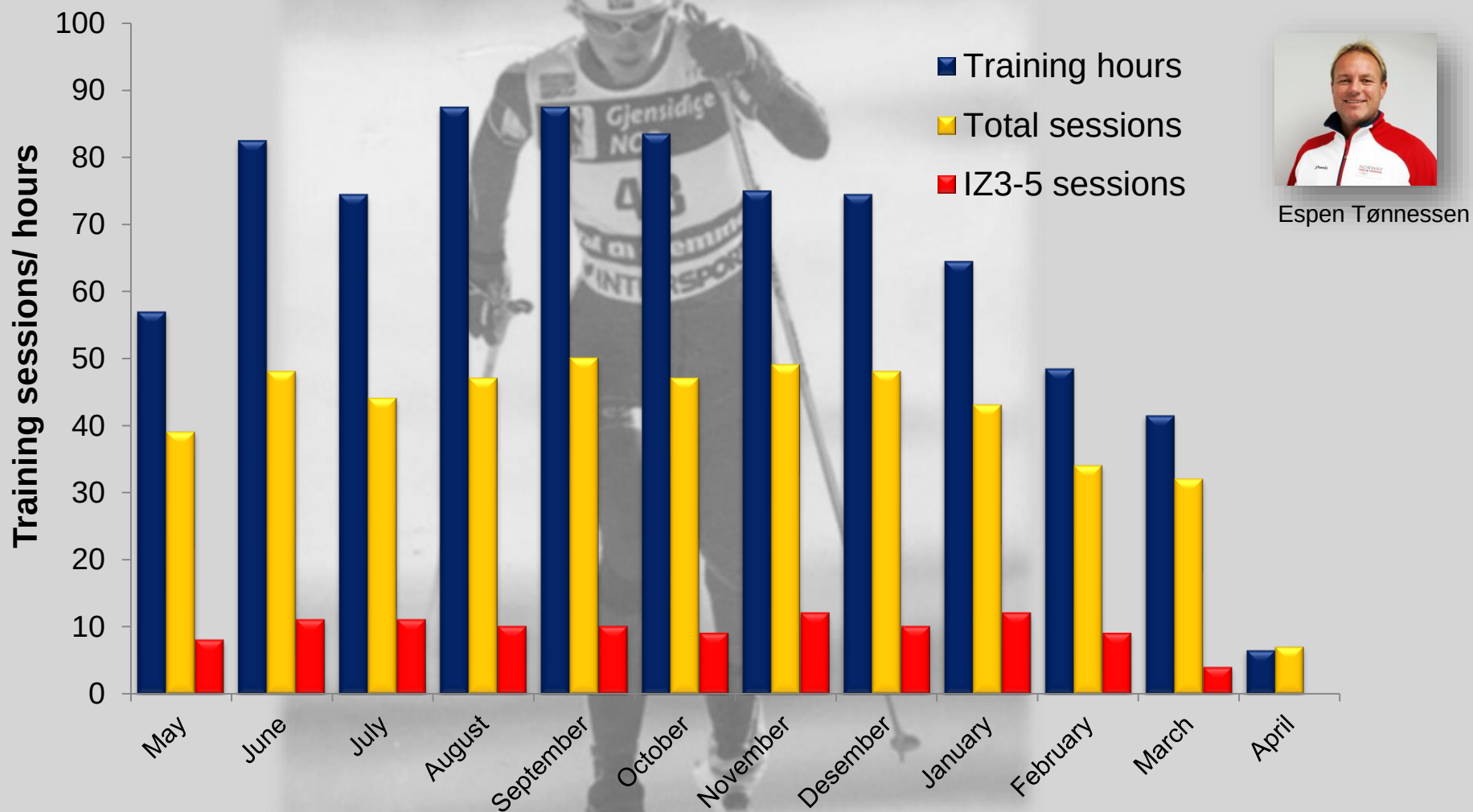


«This study supports and provides a historical context for data from elite endurance athletes suggesting that the optimal training organization for maximal performance is a **polarized model of training** with about **75%** of training performed well below the lactate threshold and **15%–20%** well above that intensity.”



Scand. J. Med. Sci. Sports.
16, 49-56, 2006.

Basic periodization of a champion XC skier



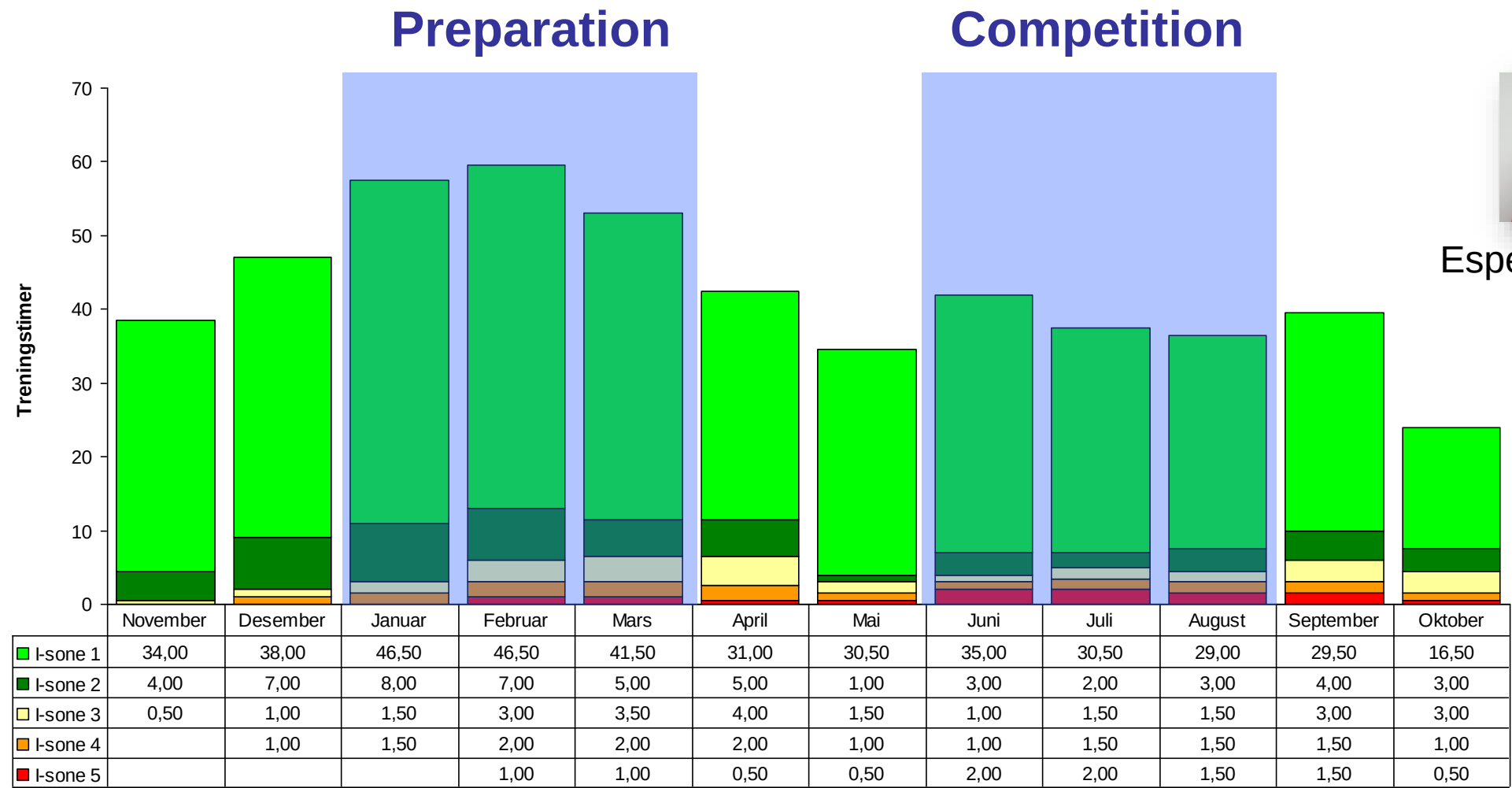
Ingrid Kristiansen

5 World Records

World Champion



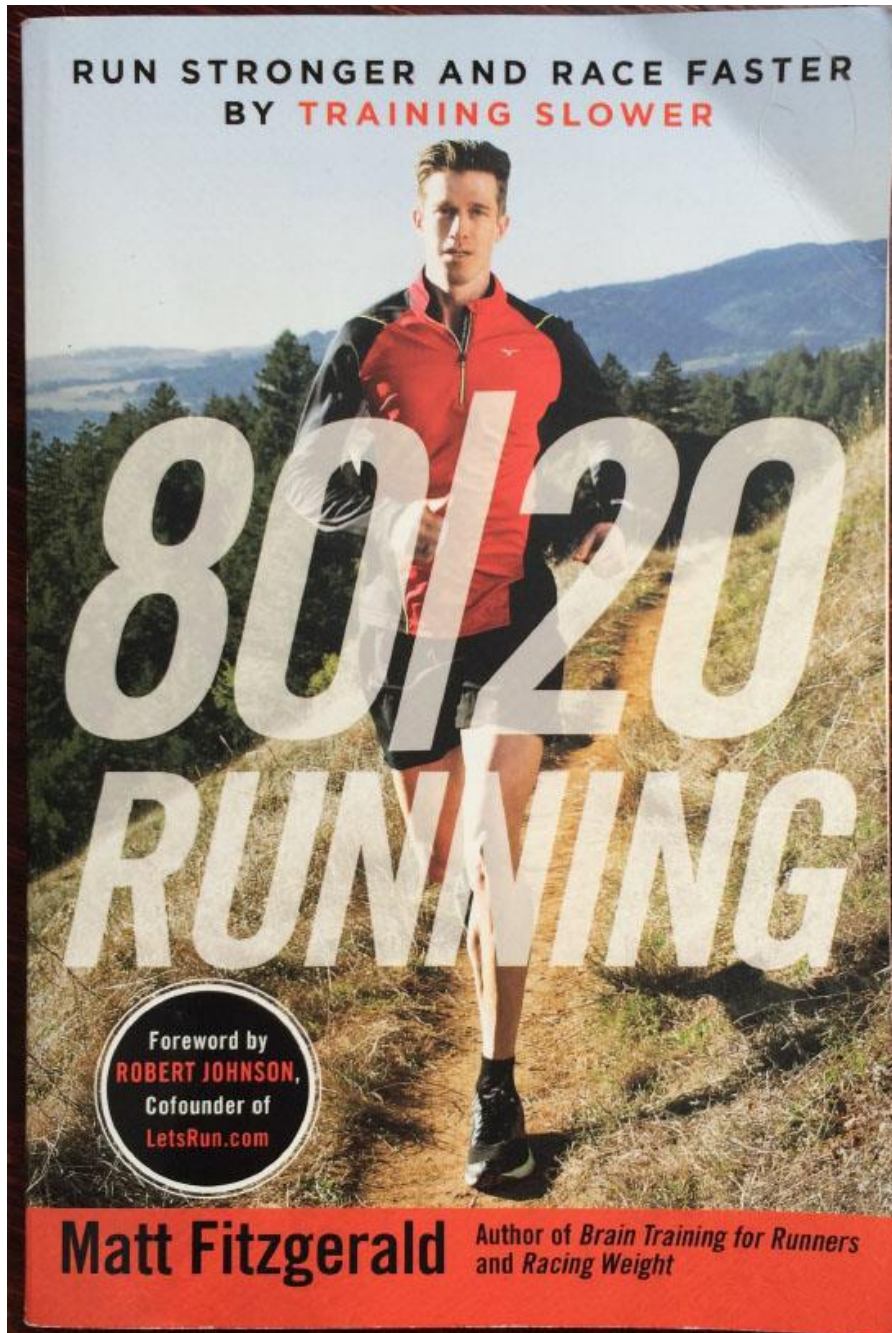
Data from Espen Tønnesen
Olympiatoppen with permission



Espen Tønnesen

and then we and others did lots more studies

- What is best practice for training intensity and duration distribution in endurance athletes?
- Intervals, Thresholds, and Long Slow Distance: the Role of Intensity and Duration in Endurance Training
- From Heart-Rate Data to Training Quantification: A Comparison of 3 Methods of Training-Intensity Analysis
- The Road to Gold: Training and Peaking Characteristics in the Year Prior to a Gold Medal Endurance Performance
- Physical and Training Characteristics of top class marathoners
- Does Polarized Training Improve Performance in Recreational Runners?
- Quantifying training intensity distribution in elite endurance athletes: Is there evidence for an "optimal" distribution?
- Training and performance characteristics among Norwegian International Rowers 1970–2001
- The Annual Training Periodization of 8 World Champions in Orienteering
- Do Elite Endurance Athletes Report Their Training Accurately?
- Training-Intensity Distribution During an Ironman Season: Relationship With Competition Performance
- **Adaptations to aerobic interval training: Interactive effects of exercise intensity and total work duration**
- Autonomic recovery after exercise in trained athletes: intensity and duration effects
- **Polarized training has greater impact on key endurance variables than threshold, high intensity, or high volume training**
- **Six weeks of a polarized training-intensity distribution leads to greater physiological and performance adaptations than a threshold model in trained cyclists**
- **A Quasi-Experimental Study of Chinese Top-Level Speed Skaters' Training Load: Threshold Versus Polarized Model**
- Impact of training intensity distribution on performance in endurance athletes
- Lactate profile changes in relation to training characteristics in junior elite cyclists
- Training methods and intensity distribution of young world-class rowers



Some of the experts whose research was discussed

***Stephen Seiler**- ok, my name is there ~45 times

Jonathon Esteve-Lanao

Veronique Billat

Samuele Marcola

Thomas Stoggl & Billy Sperlich

Stuart Galloway

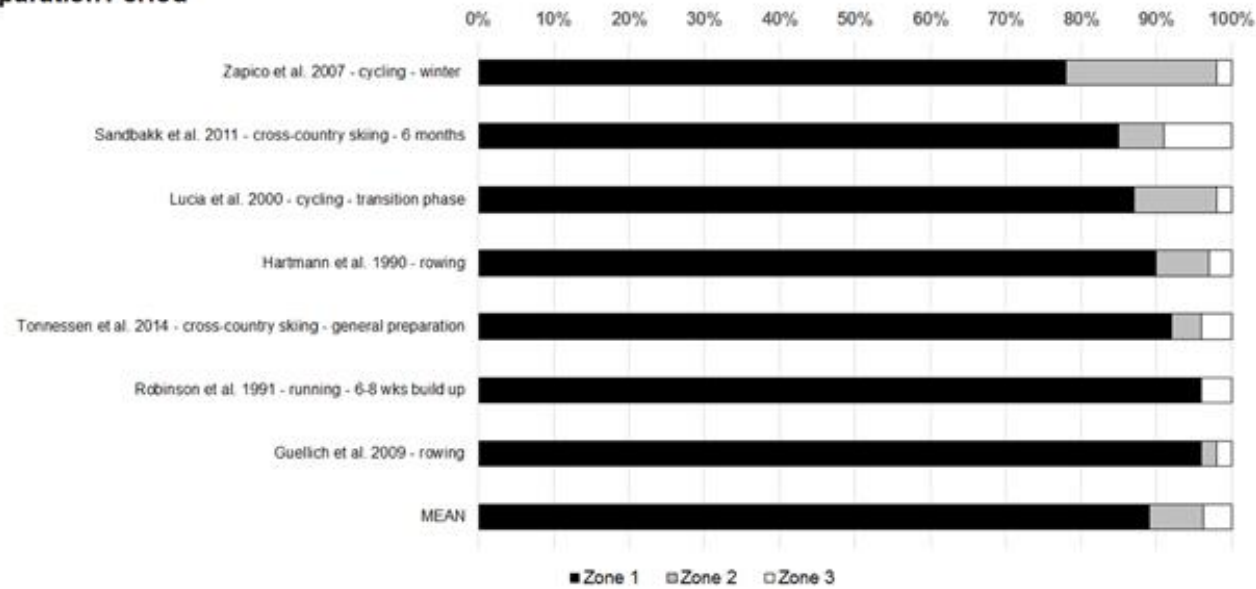
Rex Mauger & Nick Sculthorpe

«Do 80% of your running at low intensity, and the remaining 20% at moderate to high intensity. The rest is details»- Robert Johnson, founder of Letsrun.com

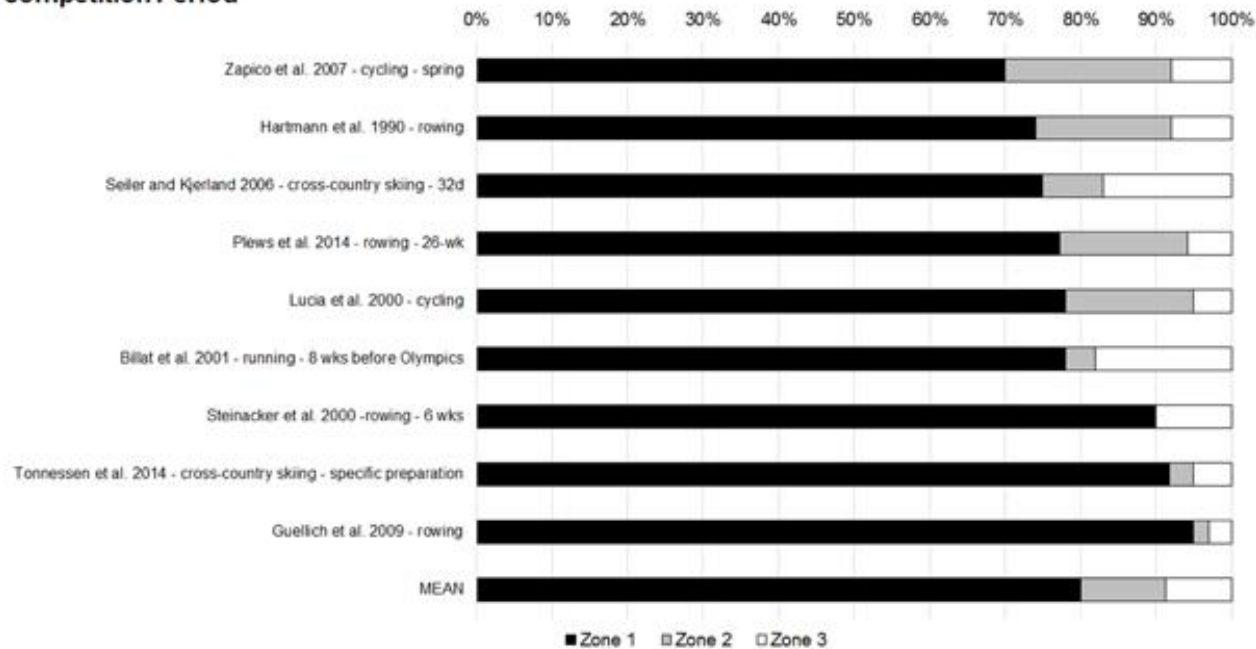
Summary of 15 years of research.....



A Preparation Period



B Pre-competition Period



FOCUSED REVIEW ARTICLE

Front. Physiol., 27 October 2015 | <http://dx.doi.org/10.3389/fphys.2015.00295>



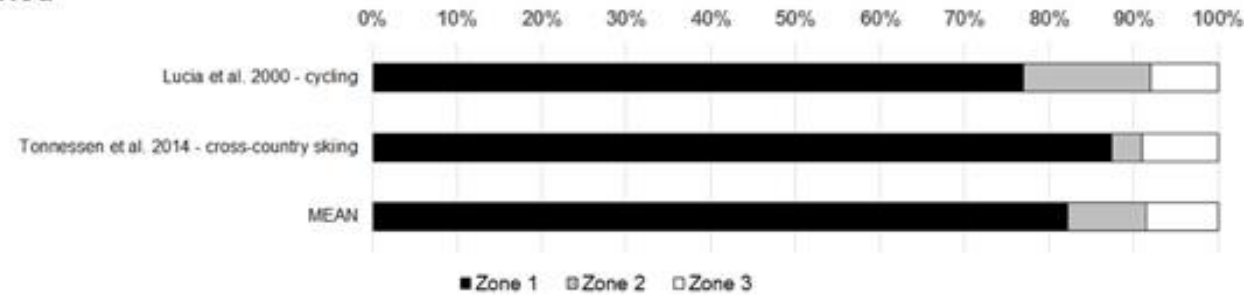
The training intensity distribution among well-trained and elite endurance athletes

Thomas L. Stöggl^{1*} and Billy Sperlich²

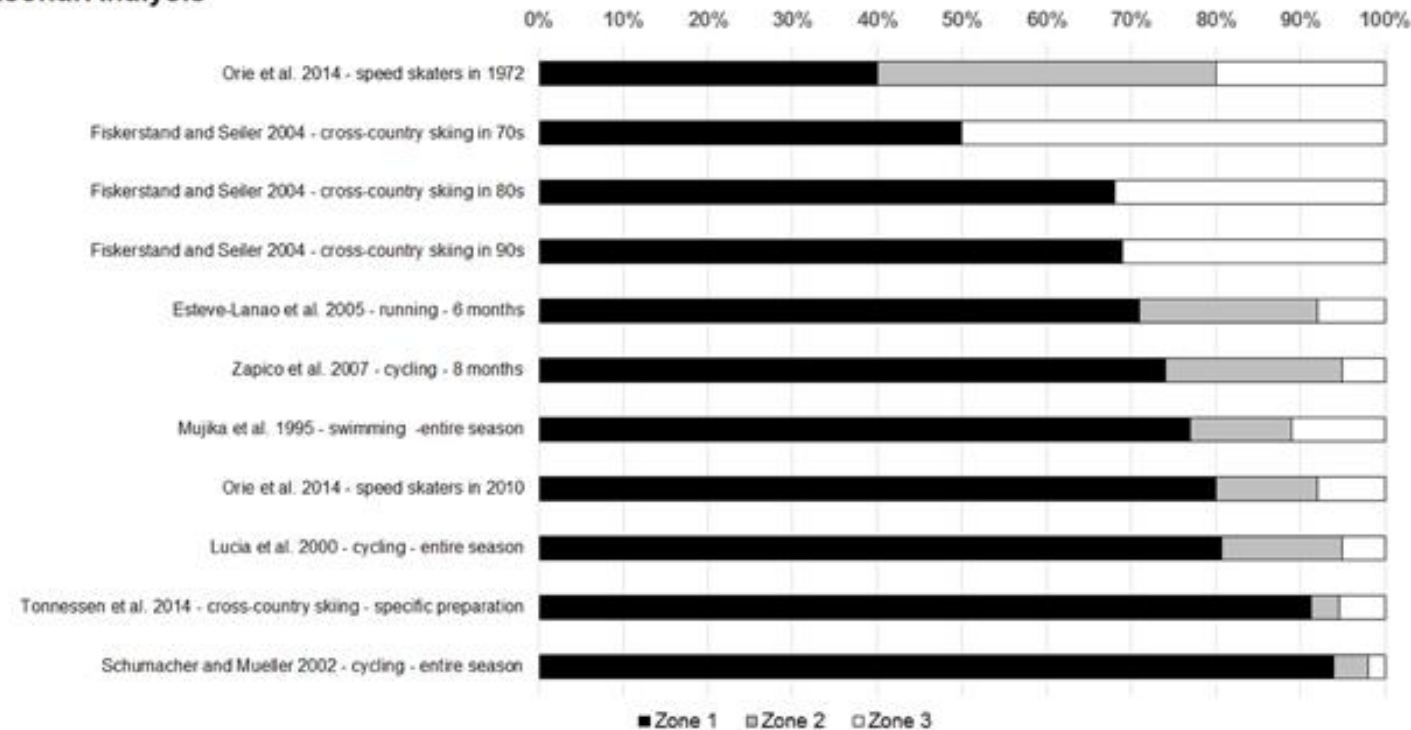
The training intensity distribution among well-trained and elite endurance athletes

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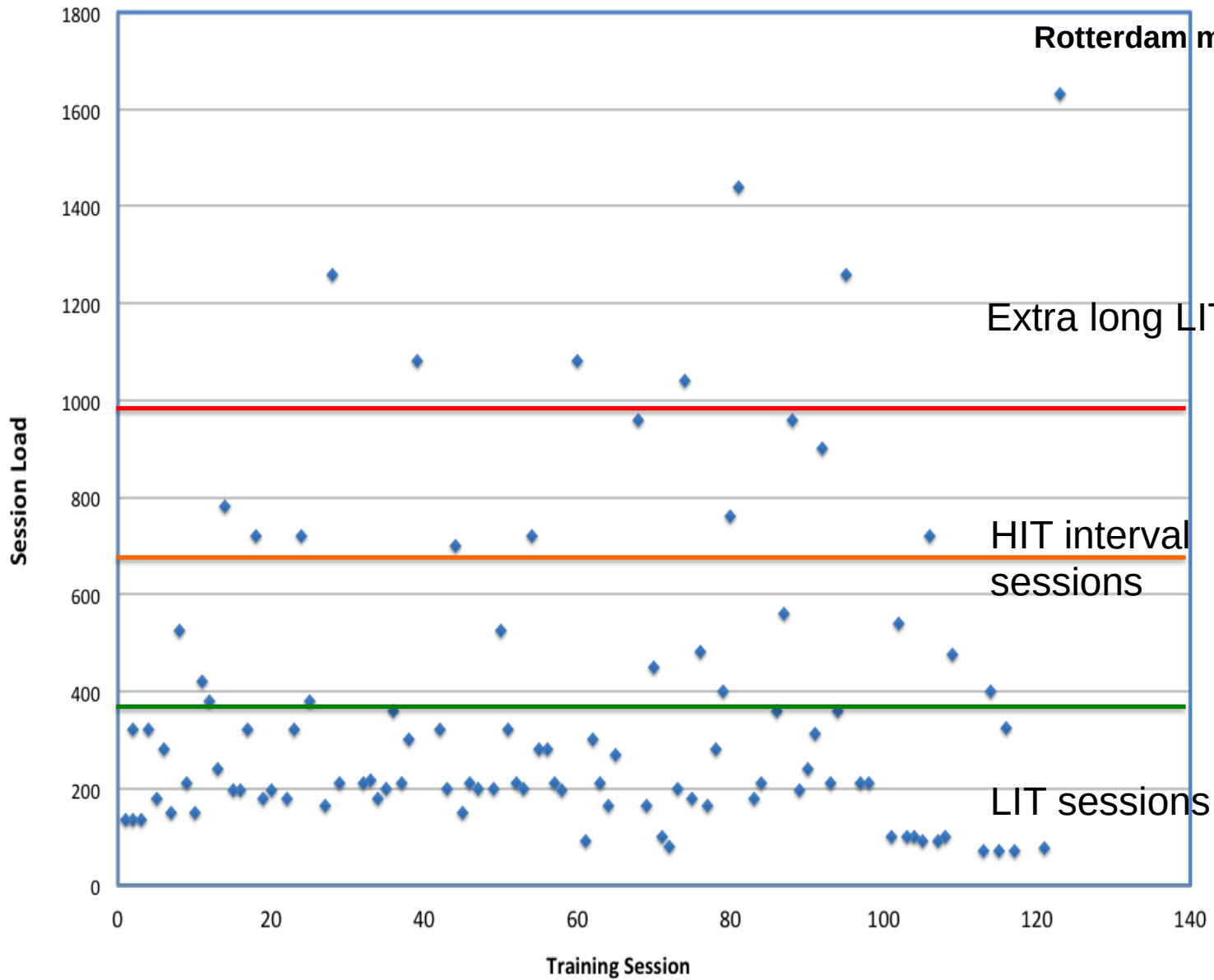
C Competition Period



D Seasonal Analysis



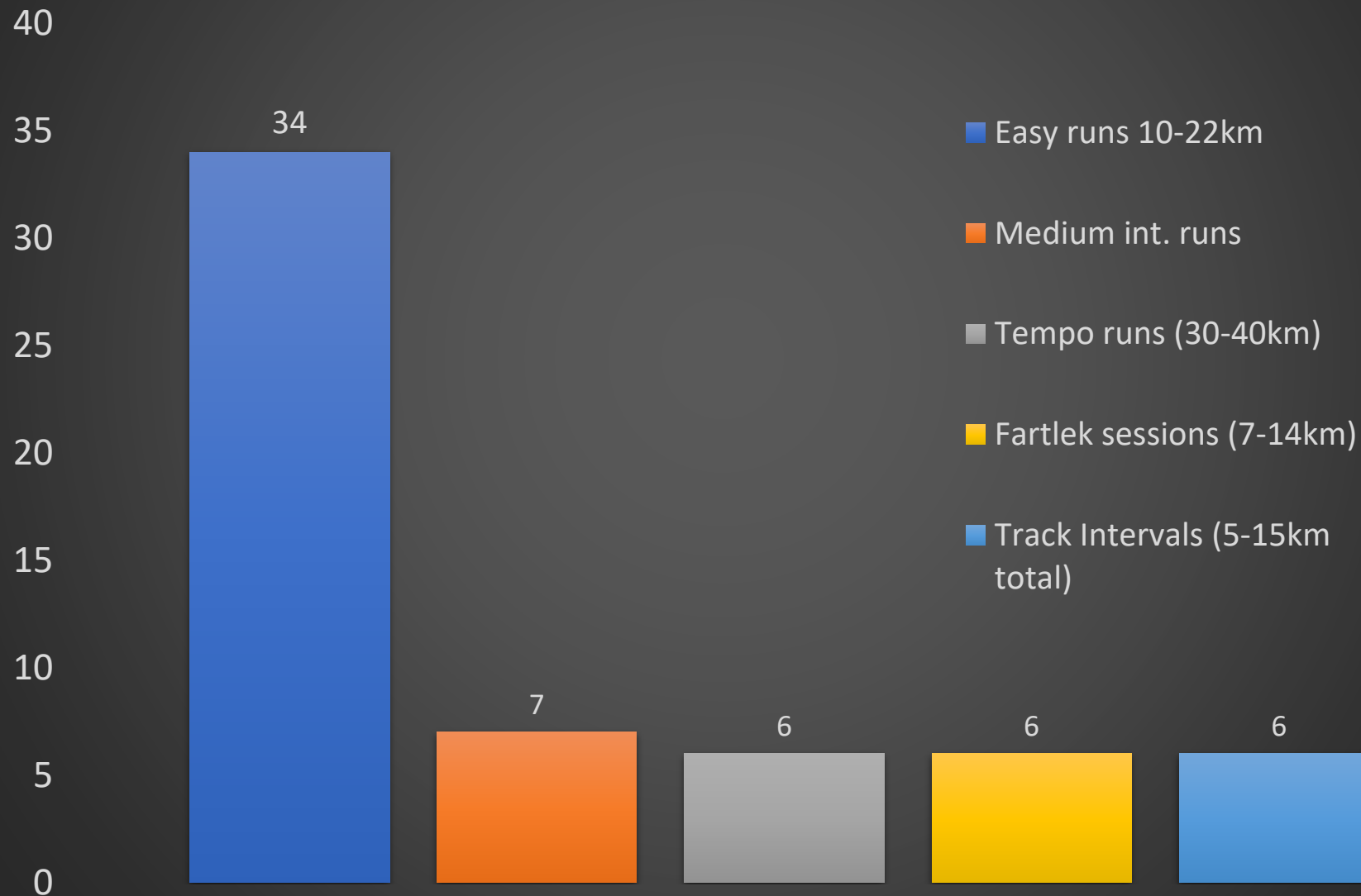
Session Loads during marathon buildup- Sylta



A 2:15 marathoner



Kipchoge- Session type distribution in 41 days (59 sessions) prior to Berlin marathon 2017



A 2:03 marathoner
(or 2:00:25)



Two «intensity poles» in the *polarized* intensity distribution model

